Protocol to SDTM in 15 mins. Is it Possible?

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Making better use of data

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Overview

At d4k, we believe clinical trial data should flow seamlessly from protocol to submission. For years, the industry has struggled with disconnected systems, manual processes, and repeated data entry. The Unified Study Definitions Model (USDM) changes this paradigm.

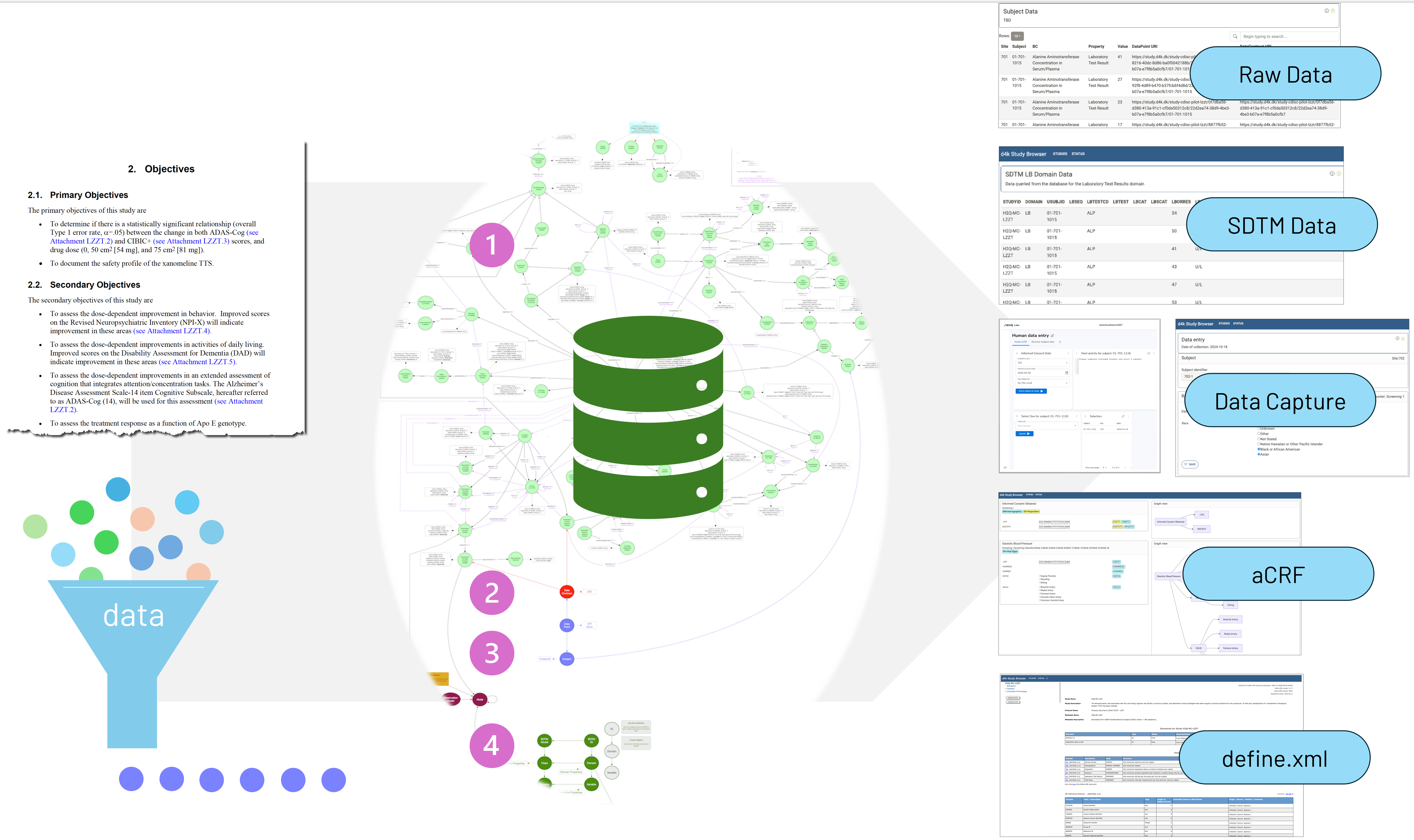
Our demonstrator combines USDM v3 (being upgraded to v4) with CDISC Biomedical Concepts and ICH M11 standards to transform protocol PDFs into SDTM datasets, CRFs, and timelines - all from a single source of truth. What once took weeks now happens in minutes, with full traceability and consistency.

Key Takeaways:

- Standards-based automation is real - USDM provides the missing link between protocol design and data collection
- One source, multiple outputs - A single model generates SDTM, CRFs, timelines, and more
- Industry transformation through collaboration - Built on CDISC standards for widespread adoption
- TransCelerate's vision realized - End-to-end digital data flow is achievable today

Start with any clinical trial protocol PDF. No special formatting required - the AI handles protocols from Phase I to Phase IV, simple to complex designs. Advanced AI extracts the necessary study design information which is automatically transformed into the USDM model, creating relationships between visits, activities, and timings. From this foundation key outputs such as aCRF specifications and define.xml can be instantly generated. Now load participant data and link with the study design context provided by USDM and SDTM datasets can be generated automatically.

In 15 minutes.



Watch it happen

See the complete transformation of a protocol PDF into analysis-ready SDTM datasets in just 15 minutes.

Watch as AI extracts the study design, automatically structures it in USDM, and generates multiple outputs - all from a single model. The same metadata that creates SDTM also produces aCRFs, define.xml, and data collection instruments.

This live demonstration proves that protocol-to-submission automation isn't theoretical - it's working today. Scan the QR code.

1 Foundation

Everything is built on top of the USDM model. The USDM provides the solid foundation for everything that follows.

2 Data Contract

Build a "data contract", the set of unique data points needed to meet the needs of the study. Allocate each datapoint a URI. The URI is the barcode for a single atomic data point, a unique identifier that persists forever. Can be used for multiple purposes: external data providers, long term retention of data ...

3 Data Load

Any data load only requires a triple of the subject identifier, the data contract URI and the data value This allows for data to be linked into the overall data, in bulk or individually

4 Link SDTM

We use a small intermediate model that links the SDTM IG via the SDTM Model to the USDM and the BCs. We take this further in that all the BCs we use are based on that same intermediate model. This allows for links to multiple versions of the IG as well as to other models such as HL7 FHIR.

Poster developed by data4knowledge ApS, for the "Transcelerate Mission Possible" Event 2025. Contact: dih@data4knowledge.dk